# Certified Krasikov–Lagarias predecessor bounds for the 3x✦1 problem to k = 22: two-sided brackets, machine-checked instances, and the finite-level loss

Daniel Palmqvist

October 2026

*Draft 0.8, 2026-10-09. All numbers are taken from RESULTS.md, certs/*/cert.json, AVENUES.md, spectrum.json, kl5.json and the literature survey reports/Collatz adjacent work and references.md.*

## Abstract

For a positive integer $a \not\equiv 0 \pmod 3$, let $\pi_a(x)$ be the number of $n \le x$ whose $3x+1$ orbit passes through $a$. Krasikov and Lagarias (Acta Arith. 109, 2003) proved $\pi_a(x) \ge x^{0.84}$ for $x \ge x_0(a)$ from a feasible point of a linear program $L^{NT}_k(\lambda)$ on the residue classes modulo $3^k$, at $k = 11$. We carry the computation to $k = 22$, where the program has $3^{21} \approx 1.05 \cdot 10^{10}$ principal variables, and obtain

$$\pi_a(x) \ge x^{\gamma} \text{ for every } \gamma < 0.92509 \text{ and } x \ge x_0(a, \gamma).$$

Every level $k = 12, \ldots, 22$ comes with an integer certificate, the float32 solution vector read as an exact integer vector, verified in int64 arithmetic by two programs that share no code; the verification of the $k = 22$ certificate (39 GB) takes four minutes. For $k = 2, \ldots, 22$ we also give infeasibility certificates, positive vectors $C$ with $F(C) < C$ strictly for the right-hand side map $F$, which prove that the program at level $k$ is infeasible slightly above the certified $\lambda$; the Krasikov–Lagarias constants $\lambda_k$ are thereby bracketed to widths between $2 \cdot 10^{-5}$ and $10^{-4}$, confirming their numerical values to four or five decimals. The levels $k = 18$, 19 and 20 are machine-checked in Lean 4 inside L. Mazur's formalisation of the Krasikov–Lagarias argument, giving theorems $x^{0.903} \le \pi_a(x)$, $x^{0.909} \le \pi_a(x)$ and $x^{0.914} \le \pi_a(x)$ for large $x$, each resting on Lean's three standard axioms and two compiled finite checks. The question whether $\lambda_k \to 2$, equivalent to $\pi_a(x) \ge x^{1-\varepsilon}$ for every $\varepsilon$, reduces to whether a single number, the mean loss $\rho_k$ from the minimum over three lifts, tends to 1. We report $\rho_k$ to $k = 22$, the spectral radius at $\lambda = 2$, and the spectrum of the linearised transfer matrix, whose gap does not close with $k$. We then show that the finite-level loss is a mean-field quantity: replacing the top-digit alignments fixed by the arithmetic of 2 and 3 by independent random ones above level 9 changes $\lambda_{15}$ by $2 \cdot 10^{-3}$ and leaves the decay ratio of $1 - \rho_k$ unchanged, so the limit question can be posed for a random model with no number theory above a fixed level. The decay ratio rises with $k$ because the distribution of the $\times 4$ weight $\lambda^{-2}\, c(4m)/c(m)$ broadens; that distribution converges with geometrically shrinking increments up to $k = 21$, which heuristically gives geometric decay of the loss and $\lambda_k \to 2$, although the ratio itself shows no deceleration yet. The same construction on $5x+1$ has the branching-walk exponent 0.6509 of Kontorovich–Lagarias as its ceiling and poses the same question. The data to $k = 22$ are consistent with $\lambda_k \to 2$ but do not establish it; we state what a proof would need.

# 1. Introduction

The 3x+1 map is T(n) = n/2 for even n and (3n+1)/2 for odd n. The Collatz conjecture asserts that every orbit reaches 1. The predecessor set of a is $\{n : T^j(n) = a \text{ for some } j \geq 0\}$; its counting function $\pi_a(x)$ measures how much of [1, x] lies in the tree below a, and the conjecture is equivalent to $\pi_1(x) = \lfloor x \rfloor$.

Lower bounds of the form $\pi_a(x) \geq x^\gamma$ have a forty-year history. Crandall (1978) showed some $\gamma > 0$ exists; Sander (1990) reached 3/10; Krasikov (1989) introduced difference inequalities on residue classes modulo $3^k$ and obtained 3/7; Wirsching (1993) 0.48; Applegate and Lagarias (1995) 0.65 by exhaustive tree search to depth 30 and 0.809 by a nonlinear program on about 20 000 variables; and Krasikov and Lagarias (2003) 0.84, from a linear program with $3^{10}$ principal variables. Lagarias's 2010 overview lists 0.84 as the record and states as open that $\pi_1(x) > c(\varepsilon)\, x^{1-\varepsilon}$ for every $\varepsilon > 0$. Krasikov and Lagarias observe in their §6 that their constants $\lambda_k$ satisfy $\lambda_k \leq \lambda_{k+1}$ and that $\lambda_k \longrightarrow 2$ would give the $x^{1-\varepsilon}$ bound; no progress on that has appeared.

Three things changed between 2003 and 2026, and this paper uses all three.

*Memory.* The level-k program has $N = 3^{k-1}$ principal variables and a right-hand side map that needs no stored index arrays: all index maps are affine in the residue. A float32 vector at k = 22 is 39 GB, which a single workstation with unified memory holds, and the map evaluates in seconds on its GPU.

*Certification.* A solution found in floating point is turned into an integer certificate and checked in exact integer arithmetic; the mathematics of the bound is the Krasikov–Lagarias theorem, and the computation reduces to a check that anyone can re-run in minutes. We add the reverse certificate: because the right-hand side map is monotone and 1-homogeneous, a vector with $F(C) < C$ strictly proves infeasibility, so $\lambda_k$ becomes a two-sided number.

*Formalisation.* Mazur's 2026 Lean 4 development replaces the analytic step of Krasikov–Lagarias (their Theorem 2.2) by a second finite certificate, an "adaptive forced potential" on the same residue classes, and checks both certificates by compiled evaluation (native_decide), so that the statement for a given k is a Lean theorem. We feed our certificates through that development for k = 18, 19 and 20.

Related work, 2024–2026. Several AI-assisted Lean projects pursue the same program; none is refereed. msharpe248 obtains $\pi_1(x) = \Omega(x^{0.8676})$ at k = 14 with a kernel-checked certificate (29 h, 37 GB) and 0.895 at k = 17 informally; Mazur obtains 0.90 and 0.901 at k = 18 via native_decide; Menta2357 reproduces the k = 11 value 0.8418 kernel-clean. At equal k these agree with ours to the precision of their respective roundings (msharpe248's values are lower at equal k because of an integer-grid weakening of the weights). On the cycle side, Barina's verification of convergence for all $n < 2^{71}$ (2025) makes Hercher's conditional result unconditional: a nontrivial cycle has at least $1.375 \cdot 10^{11}$ odd terms. Our survey of the primary sources, with provenance for every number, is in the repository.

# 2. The Krasikov–Lagarias system

## *2.1 Difference inequalities*

Fix $k \geq 2$. A residue class m mod $3^k$ with $m \equiv 2 \pmod 3$ is *principal*: its members have two predecessors under T, the even predecessor 2m and the odd predecessor $(2m - 1)/3$. Classes $\equiv 1 \pmod 3$ have only the even predecessor, and classes $\equiv 0 \pmod 3$ have only the chain $\{2^j m\}$, which contributes $O(\log x)$ to any count and is dropped. Writing $\varphi_m(y)$ for the infimum over a in class m of the number of predecessors of a not exceeding $2^y$ a, the tree structure gives $\varphi_m(y) \geq \varphi_{2m}(y - 1) + \varphi_{(2m-1)/3}(y - 1 + \alpha)$ with $\alpha = \log_2 3$, and eliminating the non-principal classes yields, for each principal m,

- $m \equiv 2 \pmod 9$: $\varphi_m(y) \geq \varphi_{4m}(y - 2) + \bar{\varphi}_{(4m-2)/3}(y + \alpha - 2)$,
- $m \equiv 5 \pmod 9$: $\varphi_m(y) \geq \varphi_{4m}(y - 2)$,

- $m \equiv 8 \pmod 9$: $\varphi_m(y) \geq \varphi_{4m}(y-2) + \bar\varphi_{(2m-1)/3}(y+\alpha-1)$,

where $\bar\varphi_r$ is the infimum over the three classes mod $3^k$ lying above the class r mod $3^{k-1}$, because the odd predecessor is known only modulo $3^{k-1}$. (The paper's equations (2.13)–(2.14) print $3^k$ where $3^{k-1}$ is meant; the reproduction of their Table 2 to seven decimals, §3.3, confirms the correction.)

### *2.2 The linear program and its map*

The ansatz $\varphi_m(y) \geq c_m \lambda^y$ turns the inequalities into the linear system $L^{NT}_k(\lambda)$: for principal m,

- $m \equiv 2 \pmod 9$: $c_m \leq \lambda^{-2} c_{4m} + \lambda^{\alpha-2} \bar c_{(4m-2)/3}$,
- $m \equiv 5 \pmod 9$: $c_m \leq \lambda^{-2} c_{4m}$,
- $m \equiv 8 \pmod 9$: $c_m \leq \lambda^{-2} c_{4m} + \lambda^{\alpha-1} \bar c_{(2m-1)/3}$,
- $c_m \geq 1$, with $\bar c_r = \min(c_r, c_{r+3^{k-1}}, c_{r+2\cdot 3^{k-1}})$.

**Theorem (Krasikov–Lagarias, Thm 2.2 and §6).** If $L^{NT}_k(\lambda)$ is feasible then for every $a \not\equiv 0 \pmod 3$ and every $\gamma < \log_2 \lambda$ there is $x_0(a,\gamma)$ with $\pi_a(x) \geq x^\gamma$ for $x \geq x_0$. The constant is explicit: $\varphi_m(y) \geq \Delta_1 c_m \lambda^y$ with $\Delta_1 = 1/(4 \max_m c_m)$.

Let $F_\lambda$ be the right-hand side map, $F(c)_m = \lambda^{-2} c_{4m} + \lambda^{\alpha-j} \bar c_{r(m)}$. F is monotone and positively 1-homogeneous on the positive cone, so it has a cone spectral radius $r(F_\lambda)$ (Collatz–Wielandt), and $L^{NT}_k(\lambda)$ is feasible iff $r(F_\lambda) \geq 1$: a feasible c satisfies $c \leq F(c)$, which forces $r \geq 1$, and conversely an eigenvector with $F(c) = rc$, $r \geq 1$, is feasible after scaling. Since r is decreasing in $\lambda$, $\lambda_k := \sup\{\lambda : L^{NT}_k(\lambda) \text{ feasible}\}$ is the unique root of $\log r(F_\lambda) = 0$.

### *2.3 Indexing*

With $N = 3^{k-1}$ and index $i \in [0, N)$ for $m = 3i + 2$, every map is affine: $4m \mapsto (4i + 2) \bmod N$; the class of m mod 9 is i mod 3 ($0 \leftrightarrow 2$, $1 \leftrightarrow 5$, $2 \leftrightarrow 8$); the auxiliary index is 4j mod N/3 for $i = 3j$ and $(2j + 1) \bmod N/3$ for $i = 3j + 2$; and $\bar c[r] = \min(c[r], c[r + N/3], c[r + 2N/3])$. The ternary digits of i are the digits of m above the lowest, and the top digit is the one the minimum ranges over. Lifting a level-(k–1) vector to level k is $c_k[i + j\,N_{k-1}] = c_{k-1}[i]$, which preserves feasibility (this is the monotonicity $\lambda_k \leq \lambda_{k+1}$).

## 3. Computation

### *3.1 Power iteration and secant search*

We compute $\lambda_k$ by a secant method on $\lambda \mapsto \log r(F_\lambda)$, with r estimated as the midpoint of the Collatz–Wielandt bounds $\min_i F(c)_i/c_i \leq r \leq \max_i F(c)_i/c_i$ after power iteration $c \leftarrow F(c)/\min F(c)$. Each $\lambda$ evaluation is warm-started from the previous iterate, and each level from the lifted solution of the level below; three to five $\lambda$ values suffice. The secant stops when its step falls below the noise level $(hi - lo)/(2|slope|)$, which is where float32 stops determining $\lambda_k$; the slope $d \log r/d\lambda \approx -0.27$ means a gap of $10^{-6}$ in r leaves $\lambda$ undetermined to about $4\cdot 10^{-6}$. Certified $\lambda$ values are rounded down to four decimals, well inside the noise.

### *3.2 Memory*

F is one Metal kernel (Apple MLX): each thread computes its indices in 64-bit integers, reads three lifts and one quadruple index, and writes one output; no index arrays exist. The normalisation 1/min c is folded into the coefficients (F is 1-homogeneous), so the iteration holds two vectors. The Collatz–Wielandt bounds are computed in chunks. For $k \geq 21$ the vector exceeds $2^{31}$ entries and is stored with shape (3, N/3) because MLX shapes are 32-bit.

When two vectors do not fit (k = 21 on 24 GB, k = 22 on 64 GB) the update is compressed: a kernel computes $\rho_i = F(c)_i/c_i$ exactly in float32 together with per-threadgroup minima and maxima, stores $d_i$

= round((ρ_i − ρ_0)/S) as int8 with ρ_0 and S from the previous iteration's bounds (S = 1.25(hi − lo)/254), and the update c ← (c/m)(ρ_0 + S d) reuses c's buffer. The quantisation perturbs the iterate by at most S/2, which is irrelevant because certification checks the final vector exactly; it raises the floor of hi − lo to about $2\cdot10^{-6}$. Peak memory is 1.26 vectors.

At k = 22 the vector (39.0 GB) exceeds Metal's single-buffer limit (38.9 GiB on the M4 Pro). We hold it as three buffers indexed by the top ternary digit, c[i] = c_j[i − j N/3]; the lift minimum becomes min(c_0[r], c_1[r], c_2[r]) and the warm start from level 21 is three copies of that vector. The split kernel agrees with the single-buffer kernel bit for bit at k ≤ 10.

### *3.3 Results*

| k | $N = 3^{k-1}$ | $\lambda_k$ (numerical) | $\lambda_{cert}$ | $\gamma_{cert} = \log_2 \lambda_{cert}$ | $\Delta\gamma$ | $\lambda_{upper}$ (infeasible) |
|---|---|---|---|---|---|---|
| 2–11 | 3 … 59 049 | Table 2 of K–L reproduced to $10^{-6}$ … $10^{-7}$ | – | 0.8418 at k = 11 | – | 1.35342 … 1.79225 |
| 12 | 177 147 | 1.8064236 | 1.8064 | 0.8531 | 0.0113 | 1.80644 |
| 13 | 531 441 | 1.8188237 | 1.8188 | 0.8630 | 0.0099 | 1.81884 |
| 14 | 1 594 323 | 1.8307723 | 1.8307 | 0.8724 | 0.0094 | 1.83079 |
| 15 | 4 782 969 | 1.8419687 | 1.8419 | 0.8812 | 0.0088 | 1.84198 |
| 16 | 14 348 907 | 1.8522346 | 1.8522 | 0.8892 | 0.0080 | 1.85225 |
| 17 | 43 046 721 | 1.8616888 | 1.8616 | 0.8965 | 0.0073 | 1.86170 |
| 18 | 129 140 163 | 1.8703250 | 1.8703 | 0.9033 | 0.0068 | 1.87034 |
| 19 | 387 420 489 | 1.8783132 | 1.8783 | 0.9094 | 0.0061 | 1.87833 |
| 20 | 1 162 261 467 | 1.8857083 | 1.8857 | 0.9151 | 0.0057 | 1.88572 |
| 21 | 3 486 784 401 | 1.8925462 | 1.8925 | 0.9203 | 0.0052 | 1.89256 |
| 22 | 10 460 353 203 | 1.8988933 | 1.8988 | 0.9251 | 0.0048 | 1.89891 |

An independent implementation (HiGHS on the explicit LP, constraints written out from residues in exact integers, no code shared with the search) reproduces $\lambda_k$ to $10^{-6}$ for k ≤ 9, and its LP solutions at $\lambda_k - 2\cdot10^{-6}$ verify exactly in rational arithmetic. Timings on an Apple M4 Pro (64 GB): k = 19, 40 s; k = 20, 151 s; k = 21, 382 s; k = 22, 110 min for three λ evaluations at 49.0 GiB peak, plus 39 min to re-tune at $\lambda_{cert}$. All vectors were produced on two machines (M5 24 GB, M4 Pro 64 GB) with $\lambda_k$ agreeing to seven decimals.

The increments Δγ fall from 0.0113 to 0.0048 with successive ratios near 0.92. A geometric fit $\gamma_k \approx \gamma_\infty - A\, r^k$ on k = 11 … 22 gives r = 0.919 and $\gamma_\infty \approx 0.980$ with maximal residual $2\cdot10^{-4}$, and predicted both $\gamma_{21}$ and $\gamma_{22}$ blind to within $10^{-4}$. We do not read a limit off it: the ratios drift upward, and models with limits 0.97 and 1 fit the same data equally well (§6).

## 4. Certificates

### *4.1 Feasibility certificates*

Let c be the float32 vector after power iteration at λ = P/Q. With $b = 23 - \lfloor \log_2 \min c \rfloor$, every $C_i = c_i\, 2^b$ is an exact integer for every positive normal float32 $c_i \geq \min c$. Choose $D = 2^d$ with d such that $\max C \cdot D < 2^{63}$ and $\max C \cdot (q + \max L_j) < 2^{63}$, proved in Python integers before the check; set $q = \lfloor D\, Q^2/P^2 \rfloor$ and $L_j = \lfloor D\, \ell_j \rfloor$ where $\ell_j$ is a rigorous rational lower bound for $\lambda^{\alpha-j}$ from mpmath interval arithmetic at 200 bits. Then check, for every i, in int64,

$$C_i\, D \leq q\, C_{(4i+2) \bmod N} + L_j\, \bar{C}_{r(i)}.$$

Since $q/D \le \lambda^{-2}$, $L_j/D \le \lambda^{\alpha-j}$ and all quantities are nonnegative, the inequality implies $C_i \le \lambda^{-2} C_{4m} + \lambda^{\alpha-j} \bar{C}$, so C is feasible for $L^{NT}_k(\lambda)$ ($C \ge 1$ by homogeneity). No rounding occurs anywhere; the only floating-point object is the input vector, whose bits define C.

Two programs perform this check. certify_fast.py forms C by bit operations (mantissa shifted by the exponent) and streams over chunks in NumPy or on the GPU in int64 (MLX int64 was verified bitwise against NumPy). verify_cert.py uses only NumPy and mpmath, forms C as float64(c)·$2^b$ followed by an exact integrality check, recomputes the interval bounds, re-derives the overflow conditions, and checks the sha256 of the vector against the certificate's metadata. A --exact mode re-checks the same integers with Python integers for small k. The certificates are rejected when tampered with (an altered entry, q + 1, L + 1, λ raised by $10^{-4}$). The check at k = 22 takes 290 s and 4 min respectively.

### *4.2 Infeasibility certificates*

**Lemma.** Let F be monotone and positively 1-homogeneous on the positive cone of $R^N$ and let $C > 0$ satisfy $F(C)_i < C_i$ for every i. Then no $d \ge 1$ satisfies $d \le F(d)$.

*Proof.* Suppose $d \ge 1$ with $d \le F(d)$. Let $s = \max_i d_i/C_i$, attained at i. *Then d ≤ sC, so by monotonicity and homogeneity d_{i} ≤ F(d)_{i*} ≤ s F(C)_{i*} < s C_{i*} = d_{i*}, a contradiction.* ■

An infeasibility certificate is therefore a vector C with the coefficients rounded the other way: $q = \lceil D Q^2/P^2 \rceil$, $L_j = \lceil D u_j \rceil$ with $u_j$ a rigorous rational upper bound for $\lambda^{\alpha-j}$, and the strict reverse inequality $q C_{4m} + L_j \bar{C} < D C_i$ checked in int64 for every i. The power-iterated vector at $\lambda_k + 10^{-5}$ serves directly (its maximal ratio is about $1 - 3\cdot10^{-6}$); at k = 3 the float32 rounding floor of the ratio gap (≈ $3\cdot10^{-5}$ at N = 9) required $\lambda_k + 2\cdot10^{-4}$, and at k = 22 the int8 path's floor required $\lambda_k + 2\cdot10^{-5}$ (1.89891; slack $1.6\cdot10^{-6}$, 2252 s to re-tune and 293 s to check). The two certificates bracket $\lambda_k$:

| k | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| width of $[\lambda_{cert}, \lambda_{upper}]$ | 2e-5 | 3e-4 | 1e-4 | 7e-5 | 7e-5 | 1e-4 | 8e-5 | 5e-5 | 4e-5 | 5e-5 |

| k | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| width | 4e-5 | 4e-5 | 9e-5 | 8e-5 | 5e-5 | 1e-4 | 4e-5 | 3e-5 | 2e-5 | 6e-5 | 1e-4 |

The brackets confirm the published Table 2 of Krasikov–Lagarias at k = 2 … 11 and show that at level k the method provably cannot give $\gamma \ge \log_2 \lambda_{upper}$; at k = 22, $\gamma_{22} \in [0.92509, 0.92517]$. The same reverse check verifies in verify_cert.py (format /2-upper).

### *4.3 Certificate metadata*

Each cert.json records k, N, λ = P/Q, D, q, $L_2$, $L_1$, b, the sha256 of the vector data and of the file, the minimum relative slack, the software versions and the claim; the vectors (1 GB at k = 18 to 39 GB at k = 22) are regenerated by the scripts or available on request. Minimal relative slacks are $6.7\cdot10^{-6}$ (k = 18), $3.1\cdot10^{-6}$ (19), $1.4\cdot10^{-6}$ (20), $1.1\cdot10^{-5}$ (21), $2.4\cdot10^{-5}$ (22); slack of order $10^{-7}$ is the float32 limit, so k ≥ 23 would need float64 polishing or a $\lambda_{cert}$ further below $\lambda_k$.

## 5. Formal verification

### *5.1 Mazur's development*

The Lean development collatz_predecessor_090 (ProofAtlas, July 2026; Lean v4.30.0-rc2, Mathlib pinned) proves, for a weights certificate W: [N) ⟶ u32 and a forced-potential certificate P: [N) ⟶ u8 on the level-k principal classes, the statement

$$\forall\, a > 0,\ a \not\equiv 0 \pmod 3\colon\ x^{\gamma} \le \pi_a(x) \text{ for all sufficiently large real } x,$$

with $\gamma = p/q$ given and $\lambda = 2^{p/q}$. W must satisfy scale · $W_i \le A \cdot W_{4i} + B_j \cdot \min(\text{lifts})$ for all i, with scale = $2^{28}$ and A, $B_j$ dyadic lower bounds for $\lambda^{-2}$, $\lambda^{\alpha-2}$, $\lambda^{\alpha-1}$ computed by exact integer root extraction (his kl_predecessor_common.py); this is our row check with the same index formulas. P must satisfy $P[4i + 2] \le P[i] + 6$, min over lifts of P at the D1 auxiliary index $\le P[3q] + 1$, and min over lifts at the D3 auxiliary index $\le P[3q + 2] - 2$; this replaces the Kőnig-type elimination in Krasikov–Lagarias's Theorem 2.2 by a finite object (his AdaptiveCriticalChoice.exponential_lower_bound). P depends on k only. Both certificates are embedded as hexadecimal chunks of 32 MiB (include_str) and checked by native_decide.

### *5.2 Our additions*

- The weights are our vector rescaled to u32 with minimum $2^{20}$ ($2^{19}$ at k = 20, where $C_{max}/C_{min}$ = 4231 would overflow 32 bits at $2^{20}$; the row check is scale-invariant). Before export the vector is re-tuned at the target $\lambda = 2^{p/q}$ by power iteration, because a vector feasible at $\lambda_1$ is not automatically feasible at $\lambda_2 < \lambda_1$: $\lambda^{\alpha-1}$ decreases with λ and the $m \equiv 8 \pmod 9$ rows can fail (our k = 18 vector at 1.8703 failed 34.8 million rows at $2^{0.903}$).
- The potential is computed as the least fixed point of the constraint system by chunked NumPy (23–26 iterations, 15 s at k = 18 to 4 min at k = 20); it reproduces Mazur's k = 18 payload bit for bit (sha256 778 584ce…).
- LoopCheck.lean replaces his list-based checker by a tail-recursive loop, proved equivalent to his Valid, which removes a List.range N (16 B per class) from the peak and makes k = 19 and 20 feasible on 64 GB.
- The level-specific modules K18, K19, K20 (principal index, power-of-two targets, source bridge, coefficient bridge, critical choice, predecessor bridge, all-targets) are generated from his K18 files by textual substitution; the generated certificate modules embed no hashes, only the chunk list, so a regenerated vector needs no Lean changes.
- Two facts about large .olean files mattered at k = 20 (9.3 GB weights, 2.3 GB potential). Lean maps each olean at a base address derived from a hash of the module name and, if that range is taken, falls back to a single read() that fails above 2 GiB on macOS. So the 9.3 GB module is imported first in every dependent module, and the potential module was renamed (Generated.K20Potential10) because the original name's range contained Mathlib.Algebra.Field.Defs; the base address of a candidate name can be predicted from the hash and checked against all existing oleans before building. The 139-chunk array also needs a higher simp recursion limit.

### *5.3 Results*

| k | γ = p/q | Lean theorem | potential check | weights check | peak RSS | axioms |
|---|---|---|---|---|---|---|
| 18 | 9032/10 000 | $x^{903/1000} \le \pi_a(x)$ (CollatzPredecessor0903) | 643 s | 871 s | 14.0 GB | 3 standard + 2 native_decide |
| 19 | 9093/10 000 | $x^{909/1000} \le \pi_a(x)$ (CollatzPredecessor0909) | 2113 s | 2554 s | 24.4 GB | 3 standard + 2 native_decide |
| 20 | 9150/10 000 | $x^{914/1000} \le \pi_a(x)$ (CollatzPredecessor0914) | 5940 s | 7382 s | 55.9 GB | 3 standard + 2 native_decide |

Each level also yields the constant-free forms _nat (natural cutoffs) and _0903x ($c \cdot x^{p/q}$ with the full certificate exponent). The publication exponent must be strictly below the certificate exponent for the constant to be absorbed (0.903 < 0.9032, 0.909 < 0.9093); at k = 20 the natural choice 915/1000 equals 9150/10 000, so the unit-coefficient theorem is stated at 914/1000 and the constant form keeps 9150/10 000. #print axioms lists propext, Classical.choice, Quot.sound and the two named native_decide axioms per level, the same trust base as Mazur's 0.90. Mazur's own Python verifier passes on the three

exports (129 140 163, 387 420 489 and 1 162 261 467 rows, zero failures), and his potential verifier passes on the k = 19 and k = 20 potentials (645 700 815 and 1 937 102 445 inequalities).

### *5.4 Trust*

native_decide places the Lean compiler in the trust base. A kernel soundness bug (Lean issue #14 576, July 2026, fixed in 4.32.2) was exposed by an AI-assisted "disproof" of the Collatz conjecture; we therefore report the axiom audit of every theorem, keep both external verifiers in the loop, and recommend re-running the builds on a toolchain at or above 4.32.2. The memory of native_decide scales linearly with N (about 8–24 bytes per class), so k = 21 and 22 (estimated 90 and 270 GiB) are out of reach of this route on 64 GB; a streamed verified checker in the style of $\text{cake}_{\text{lpr}}$ or lrat-catcher, proved once and run natively in constant memory, is the natural next step and would also move the compiled code out of the trust base of each individual certificate.

## 6. The finite-level loss and the question $\lambda_k \to 2$

### *6.1 One number*

Summing the inequalities over all principal classes, and using that $m \mapsto 4m$ is a bijection of the principal classes and that each auxiliary map is a bijection from a third of them onto the auxiliary classes, gives

$$(1 - \lambda^{-2})\, \bar{c}_{k,k} \le (1/3)(\lambda^{\alpha-1} + \lambda^{\alpha-2})\, \bar{c}_{k-1,k},$$

with $\bar{c}_{k,k}$ the mean of c and $\bar{c}_{k-1,k}$ the mean of $\bar{c}$. Write $\rho_k = \bar{c}_{k-1,k}/\bar{c}_{k,k} \le 1$ for the mean loss from the minimum over lifts. At $\lambda = 2$ both sides equal 3/4 when $\rho = 1$, so $\lambda_k \to 2 \iff \rho_k \to 1$. On all our solutions the summed inequality is tight to $10^{-5}$ (the difference is $\lambda_{\text{cert}} < \lambda_k$), so $\lambda_k$ is, to that precision, a function of $\rho_k$ alone: $1 - \lambda^{-2} = (1/3)(\lambda^{\alpha-1} + \lambda^{\alpha-2})\, \rho$.

| k | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $1 - \rho_k$ | 0.0573 | 0.0524 | 0.0482 | 0.0443 | 0.0407 | 0.0375 | 0.0347 | 0.0321 | 0.0298 | 0.0277 | 0.0258 | 0.0241 |
| ratio | 0.913 | 0.913 | 0.920 | 0.919 | 0.920 | 0.922 | 0.924 | 0.927 | 0.928 | 0.930 | 0.931 | 0.933 |

The ratio rises monotonically, so $1 - \rho_k$ is not geometric. Geometric-with-offset (limit $\rho\infty = 0.991$, $\gamma\infty \approx 0.971$), stretched-exponential ($\rho\infty = 1$) and power-law ($\rho\infty = 1$) models fit k = 11 … 22 with the same maximal residual $\approx 10^{-4}$. The data cannot decide.

### *6.2 The λ = 2 operator*

At $\lambda = 2$ the same map $F_2$ has spectral radius $r_k < 1$, non-decreasing in k with limit $r\infty \le 1$, and $\lambda_k \to 2 \iff r\infty = 1$. The exact identity $r_k = 1/4 + (3/4)\rho'_k$ holds, where $\rho'_k$ is the mixing ratio of the $\lambda = 2$ Perron vector itself (checked to 7 decimals at every k up to 22). The sequence $1 - r_k$ for k = 5 … 22 shrinks by ratios rising from 0.86 to 0.924; at k = 22, $r_{22} = 0.974971$ and $\rho'_{22} = 0.966628$. Geometric-with-offset (limit $1 - r\infty = 0.0072$) and stretched-exponential (limit 0) fits reach maximal residual $1.8 \cdot 10^{-4}$ on k = 5 … 22; fitted on $k \le 16$ the offset model misses k = 22 by $-4.3 \cdot 10^{-4}$ and the stretched model by $-7.5 \cdot 10^{-4}$, pure geometric by $-1.5 \cdot 10^{-3}$.

### *6.3 A renormalisation reading and a prediction*

Lift the level-k solution to level k + 1 (its top-digit sensitivity is then 0) and apply F once: the injected top-digit sensitivity is exactly $(3/4)s_k$ at $\lambda = 2$, because the mean is preserved and only the auxiliary term, whose mass fraction is 3/4, selects lifts by the new digit. Iterating to the fixed point amplifies it by $A_k = s_{k+1}/((3/4)s_k) = 1.20 \to 1.23$ for k = 8 … 21, and the per-level ratio of $1 - r_k$ is $(3/4)A_k$. Hence $\lambda_k \to 2$ iff $A\infty < 4/3$. The amplification comes from the ×4 term recirculating top-digit variation with weight 1/4; if a fraction e of that variation survives the minimum, $A = 1/(1 - e/4)$. Writing the fixed point as c

= P + Q with P the ×4 term (mass fraction 0.275 ⟶ 0.262, tending to 1/4) and Q the auxiliary term, the surviving fraction is $e_k$ = 0.614, 0.666, 0.692, 0.705, 0.715, 0.724, 0.731, 0.737, 0.742, 0.746 for k = 8 … 17, with shrinking increments; and the argmins of P and Q over the top digit agree on 0.332–0.333 of the classes at every k, exactly as if independent. If $e_\infty \approx 0.78$–$0.80$ the per-level ratio saturates near $(3/4)/(1 - e_\infty/4) \approx 0.935$–$0.94$, and $1 - r_k$ eventually decays geometrically: $\lambda_k \longrightarrow 2$.

The prediction is therefore that the ratios keep rising but level off near 0.94 rather than approaching 1. k = 22 gives 0.924, still rising by about 0.002 per level, consistent and not decisive; at the current rate 0.94 would be reached near k = 31. What a proof would need is that the two argmins stay independent and the two variations stay comparable in the limit measure.

### *6.4 The limit object*

The total variation between the level-j marginals of the normalised solutions at consecutive k decays geometrically (ratio ≈ 0.92), so a limit measure on $Z_3$ exists. It is singular: $C_{max}/C_{min}$ grows like $1.52^k$ (147 at k = 12, 9851 at k = 22), the mass in the top 1 % of classes rises from 6.5 % to over 9 %, and the Hölder exponent of the digit profile on coarse scales falls from 0.14 (k = 12) to 0.085 (k = 19), extrapolating to 0.04–0.09, possibly 0. The boundary layer is universal: the Hölder exponent at distance d from the top digit is 0.665, 0.401, 0.293, 0.232, 0.194 for d = 1 … 5, identical for every k to three decimals. To first order it is not a new object. Let R(c) be c minimised over its top ternary digit and renormalised to mean 1, a vector on level k − 1. Then $R(c_k)$ is close to $c_{k-1}$: the total variation distance is 0.0126 at k = 8 and 0.0045 at k = 17, four times smaller than the distance from $c_k$ to the lift of $c_{k-1}$, and $R^2(c_k)$ is likewise close to $c_{k-2}$. The oscillation over a ball with d free top digits is therefore the loss accumulated along the chain $c_k \longrightarrow c_{k-1} \longrightarrow \cdots$, $\mathrm{osc}_k(d) \approx \Sigma_{i<d}\,(1 - \rho_{k-i})$: at k = 17 the measured values 0.0347, 0.0720, 0.1119, 0.1543, 0.1991 stand against the partial sums 0.0347, 0.0722, 0.1129, 0.1571, 0.2053. The exponents $\beta(d) = \log_3(S(d+1)/S(d))$ of the partial sums are 0.668, 0.407, 0.301, 0.243, within 0.003–0.011 of the observed ones; with $1 - \rho_k \approx q^k$ they are logarithms of ratios of geometric partial sums, $\log_3 2$, $\log_3 3/2$, $\log_3 4/3$, … pulled up slightly by q ≈ 0.92. The five numbers carry no information beyond the ratios of successive $1 - \rho_k$. Two remarks. The observed exponents are more k-independent than the model, which drifts by 0.005 over k = 11 … 17, so a second-order effect remains. And $R(c_k)$ is not feasible at level k − 1, because the minimum of a sum exceeds the sum of the minima; the defect $R(F_k\, c) - F_{k-1}(R\, c)$ is the misalignment of §6.3, and its size is what decides whether $1 - \rho_k \longrightarrow 0$ (boundary_layer.py). Integrating the limit inequality on $X = \{m \in Z_3 : m \equiv 2 \pmod 3\}$ with Haar measure gives $1 \le \lambda^{-2} + (1/3)(\lambda^{\alpha-2} + \lambda^{\alpha-1})$ with equality exactly at λ = 2, so λ = 2 is the critical value of the limit problem and the finite-k deficit is precisely $1 - \rho_k$.

### *6.5 The finite-level loss is mean-field*

The object left open in §6.4 is the defect $D_k = R(F_k\, c_k) - F_{k-1}(R\, c_k)$. We measured it (defect.py), tested the independence it suggests by randomising the system (meanfield.py), and traced the rise of the decay ratio $q_k = (1 - \rho_k)/(1 - \rho_{k-1})$ to a measurable distribution (weights.py, popdyn.py). The conclusion of this section is a heuristic and is marked as such.

**The defect.** The three level-k rows above a level-(k − 1) index r use the three lifts of the ×4 target and the three lifts of the auxiliary target in a paired order, so $D_k \ge 0$ pointwise, with equality exactly where the two argmins agree (and on the class 5 mod 9, which has no auxiliary term). Because $F_k\, c_k = c_k$, the auxiliary vector $\bar c = R\, c_k$ is a strict super-solution at level k − 1, $\bar c = F_{k-1}(\bar c) + D_k$, and the level-(k − 1) summation identity gives the exact relation

$$\text{mean } D_k \,/\, \text{mean } \bar c = g(\lambda_k)\,(\rho_k - \rho'_k), \quad g(\lambda) = (\lambda^{\alpha-1} + \lambda^{\alpha-2})/3, \quad \rho'_k = \text{mean}(R^2\, c_k)/\text{mean}(R\, c_k),$$

which the data satisfy to all printed digits for k ≤ 19. The mean defect is therefore bookkeeping: with $e_k = \rho_{k-1} - \rho'_k$ it equals $g(1 - \rho_k)[(1 - q_k)/q_k + e_k/(1 - \rho_k)]$, and the question $\lambda_k \longrightarrow 2$ is unchanged, $q_k \longrightarrow q_\infty < 1$ or $q_k \longrightarrow 1$. We find $q_k$ = 0.913 at k = 12 rising to 0.931 at k = 21, linear in k to within 0.003 (slope

0.0018 per level, no measurable curvature), and $e_k/(1 - \rho_k)$ falling from 0.048 to 0.029. The pointwise structure is cleaner than the means. The two argmins are independent: they agree with probability 0.332 for $k \geq 13$, both marginals are uniform to 0.001, the joint law differs from the product by at most 0.002, and which lift wins is unpredictable from the lowest four digits. The expected size of the defect, by contrast, is a coarse property: its conditional mean given the top digit is uniform to $10^{-3}$, the spread over a single digit falls from 0.60 of the mean at digit 1 to below 0.06 from digit 4, and the top six digits explain 0.04 % of its variance. The excess splits 64 % ×4 term and 36 % auxiliary term at every k. The defect is thus the product of two independent fine-scale choices whose expected size is fixed by the residue mod $3^4$, which is the mean-field structure that §6.3 assumed.

**Random alignments reproduce $\lambda_k$.** The index maps are recursive in the top ternary digit: writing a level-k index as $i_{low} + t \cdot N/3$, the ×4 target is $nxt_{k-1}(i_{low}) + \sigma(t) \cdot N/3$ and the auxiliary target is $aux_{k-1}(i_{low}) + \pi(t) \cdot N/9$, where $\sigma$ is a cyclic shift and $\pi$ a shift or a reflection of {0, 1, 2}, both fixed by $i_{low}$ through the arithmetic of 2 and 3. We replaced $\sigma$ and $\pi$ by independent uniform random bijections above a chosen level and recomputed $\lambda_k$ and $\rho_k$; nxt stays a bijection and aux stays 3-to-1, so the summed identity and the ceiling $\lambda = 2$ are unchanged.

| randomised levels | change in $\lambda_k$ | change in $1 - \rho_k$ | $q_k$ |
|---|---|---|---|
| top level only, k = 9 … 13 (8 seeds) | within $\pm 3 \cdot 10^{-4}$ | +0.3 % | unchanged |
| 9 … k at k = 14 and k = 15 (6 seeds) | $-2.1 \cdot 10^{-3} \pm 5 \cdot 10^{-4}$ at both | +1.6 % at both | unchanged within 0.005 |
| 12 … 15 | $-7.8 \cdot 10^{-4} \pm 9 \cdot 10^{-5}$ | +0.6 % | within 0.003 |
| 6 … 13 | $-0.9 \cdot 10^{-2}$ | +6 % | noisy |
| 3 … 11 | $-4 \cdot 10^{-2}$ | +27 % | – |

The cost of randomising the fine levels saturates at 15 % of one level's gain instead of growing with the number of levels, so the per-level increments of $\lambda_k$ agree from k = 14 on; the coarse levels carry the arithmetic. The question $\lambda_k \longrightarrow 2$ can therefore be posed for a random Krasikov–Lagarias model, the exact system up to a fixed level lifted with independent uniform alignments above it, in which it becomes a quenched-versus-annealed question of the kind studied for branching random walks. The exact system is slightly better than random and no comparison theorem is known.

**Why $q_k$ rises.** The level-(k + 1) triple above a level-k class m is a $\lambda^{-2}$-weighted copy of the triple above 4m plus an injected triple from the auxiliary target, so the contraction of the loss from one level to the next is governed by the ×4 weight $a_m = \lambda^{-2}\, c(4m)/c(m)$, which equals 1 on class 5 and leaves $1 - a_m$ for the injection. A class with $a_m$ near 1 transports its loss without contraction; one with $a_m$ near 0 is dominated by the injection. A broader distribution of $a_m$ means a weaker contraction and a larger $q_k$, and the distribution does broaden:

| k | 8 | 12 | 16 | 19 | 21 | fitted limit (ratio of increments) |
|---|---|---|---|---|---|---|
| std of $a_m$ within class 2 mod 9 | 0.227 | 0.244 | 0.255 | 0.260 | 0.264 | 0.278 (0.91) |
| std of $a_m$ within class 8 mod 9 | 0.119 | 0.127 | 0.132 | 0.135 | 0.137 | 0.145 (0.92) |
| std of log c | 0.714 | 0.798 | 0.859 | 0.894 | 0.913 | 1.026 (0.93) |

The increments shrink geometrically (maximal residuals of the fits $10^{-4}$ to $6 \cdot 10^{-4}$), so the weight distribution converges, while the growth $C_{max}/C_{min} \propto 1.52^k$ is confined to the extreme tail. The naive variance recursion $q^2 = E[(1 - a)^2]/(1 - E[a^2])$ gives 0.85 at k = 8 rising to 0.89 at k = 21 (magnitude weighted), well below the actual 0.91 to 0.93 because the minimum is not additive, but rising at the same rate as the data. A cavity model of the random system, with sibling triples typed by the level-8

residue (popdyn.py), reproduces this mechanism but cannot decide the limit: the loss factor $\bar{c}_{k+1}/c_k$ fluctuates by one third of the loss in the exact system, with sibling correlation 0.35, and different closures of that fluctuation move q at k = 60 between 0.95 and 0.99.

**Heuristic conclusion.** If the distribution of the ×4 weight converges, the contraction per level converges to a constant below 1, so $q_k \to q_\infty < 1$, $1 - \rho_k \to 0$ geometrically and $\lambda_k \to 2$; the fitted limits put $q_\infty$ near 0.95. Against this, $q_k$ itself is linear in k over k = 12 … 21 with no measurable curvature, so the deceleration the mechanism predicts is not yet visible. We state this as an expectation, not a result. A proof along these lines needs two statements about the limit object: that the fine alignments may be treated as independent, which the scrambling experiment supports to $2{\cdot}10^{-3}$ in λ, and that the distribution of the ×4 weight converges, which the data to k = 21 support with geometrically shrinking increments. Together they would give a contraction estimate for the loss.

### *6.6 The spectrum of the linearisation*

At the Perron vector the minimum in every row is attained by a fixed lift, so F is a sparse matrix J there, with $J[i, 4i] = \lambda^{-2}$ and $J[i, \text{argmin lift}] = \lambda^{\alpha-j}$. We computed its spectrum (dense to k = 9, Arnoldi top-40 to k = 15) at $\lambda = \lambda_k$ and $\lambda = 2$.

| k | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|
| $\lvert\lambda_2\rvert$ at $\lambda_k$ | 0.857 | 0.865 | 0.861 | 0.865 | 0.861 | 0.865 | 0.867 | 0.865 |
| eigenvalues within 1 % of $\lvert\lambda_2\rvert$ (of top 40) | 4 | 4 | 6 | 6 | 20 | 27 | 25 | 39 |
| $\lvert\lambda_2\rvert/\lambda_1$ at $\lambda = 2$ | 0.884 | 0.884 | 0.887 | 0.880 | 0.892 | 0.880 | 0.874 | 0.876 |

The gap does not close: $|\lambda_2|$ is pinned at 0.86–0.87 while a ring of eigenvalues fills in at that radius, the finite-k shadow of the edge of the essential spectrum. The second eigenvector is delocalised: its energy is 0.667 on every ternary digit (the value for a vector with no digit structure) and 0.71 on the top digit, while the Perron vector's digit profile decays geometrically from 0.38 to 0.004; $\lambda_2$ is complex with a rotating argument. For $k \le 9$ the full spectrum is a disc of radius ≈ 0.8 containing 99 % of the eigenvalues, a thin ring at 0.865, and the isolated Perron root. Consequently the k-dependence of $1 - \rho_k$, and the slowing of the power iteration at large k (effective contraction ≈ 0.97 at k = 22), are not set by the linear mixing rate; they come from the minimum, that is, from argmin switching as the iterate moves and from non-normal transients. The question $\lambda_k \to 2$ is a question about the minimum structure, which is what §6.3 addresses, not about the spectrum of the linearisation. The structural reason is that the doubling branch $m \mapsto 4m$ is an isometry of $Z_3$ and mixes nothing: on Hölder-α functions the essential radius should be near $1/4 + (3/4)3^{-\alpha}$, which approaches 1 as $\alpha \to 0$, and the limit vector's Hölder exponent is at most 0.09.

### *6.7 The same construction on 5x+1*

The same construction applies to T(n) = n/2, (5n+1)/2: principal classes $a \equiv 3 \pmod 5$ (odd predecessor (2a − 1)/5), four doublings return to a principal class ($\lambda^{-4}\, c_{16a}$), and the odd predecessor falls into one of five cases by a mod 25 (weight $\lambda^{\beta-1-d}$ with $\beta = \log_2 5$ and d = 0, 1, 2, 3 further doublings, or dropped when it is $\equiv 0 \bmod 5$), with the minimum over five lifts. A residue-walking LP (HiGHS, constraints generated by following the map on residues, no index formulas) reproduces Table 2 for 3x+1 to $10^{-6}$ and matches the 5x+1 index formulas to $10^{-8}$ for $k \le 6$; a Metal kernel matches the NumPy map to $10^{-7}$.

Summing the 5x+1 inequalities gives $(1 - \lambda^{-4})\,\bar{c} \le (\lambda^{\beta}/5)(\lambda^{-1} + \lambda^{-2} + \lambda^{-3} + \lambda^{-4})\,\rho\,\bar{c}$ with $\rho \le 1$ the mean loss from the minimum over five lifts. At ρ = 1 the two sides agree at λ = 2, the same mass-conservation identity as for 3x+1, but that is not the binding root: they also cross at $\lambda^* = 1.5701677$, and every λ in (λ*, 2) violates the summed inequality even with ρ = 1. So $\lambda_k(5) \le \lambda^*$ for every k. Factoring $1 - \lambda^{-4} = (1 - \lambda^{-1})(1 + \lambda^{-1} + \lambda^{-2} + \lambda^{-3})$ reduces the ρ = 1 equation to $1/\lambda + \lambda^{\beta-1}/5 = 1$, the Malthusian equation of the naive backward branching walk (child 2a at height +1; child (2a − 1)/5 at height 1 − β with probability 1/5).

Its root $\gamma^* = \log_2 \lambda^* = 0.650919$ is the exponent $\eta_{5,\mathrm{BP}}$ of Kontorovich–Lagarias (Theorem 8.10) to every printed digit; the 0.649 of their Conjecture 7.2 is the same number rounded, and Volkov's 0.678 lies above the ceiling, beyond the reach of this method at any k. For 3x+1 the same factoring gives $1/\lambda + \lambda^{\alpha-1}/3 = 1$, whose only root above 1 is 2. In both systems, therefore,

$$\lambda_k \longrightarrow (\text{Malthusian root of the branching model}) \iff \rho_k \longrightarrow 1.$$

5x+1 is thus not a calibration case with a known answer. The heuristic prediction for it is the ceiling itself, $1 - \rho_k(5) \longrightarrow 0$, exactly as for 3x+1; it is a second instance of the open question.

| k | 6 | 8 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|
| $\gamma_k(5)$ | 0.411 | 0.449 | 0.477 | 0.489 | 0.499 | 0.508 | 0.517 |
| $1 - \rho_k(5)$ | 0.150 | 0.113 | 0.090 | 0.082 | 0.074 | 0.068 | 0.063 |
| ratio | 0.83 | 0.88 | 0.90 | 0.90 | 0.91 | 0.915 | 0.919 |

At k = 14 the gap to the ceiling is 0.134 in γ, close to the 3x+1 gap of 0.128 at the same k, and the ratios of $1 - \rho_k(5)$ rise from 0.83 to 0.919, faster than for 3x+1. What 5x+1 offers is a second sequence to set beside the first under the same conjectural answer, and a constraint on proofs: an argument that $\rho_k \longrightarrow 1$ must work for 5x+1 as well, which excludes arguments that use the contracting drift of 3x+1 (the 5x+1 walk drifts upward, since $\log_2 5 > 2$). Equivalently, $\rho_k \longrightarrow 1$ says that the deterministic residue system realises the annealed, first-moment growth rate of the branching walk. A calibration against a known negative would need a map whose deterministic predecessor exponent is provably below its Malthusian root; we know of none. (No explicit predecessor-count exponent for 5x+1 exists in the literature; $\pi_a(x) \geq x^{0.516}$ numerically at k = 14 appears to be the first, pending the same certification.)

## 7. What is and is not established

All results here are density statements, and even the limit $\gamma \longrightarrow 1$ would exclude neither a nontrivial cycle nor a divergent orbit. Cycles are the province of Diophantine approximation: with Barina's verification to $2^{71}$ and Hercher's analysis, a nontrivial cycle has at least $1.375 \cdot 10^{11}$ odd terms and period at least $2.18 \cdot 10^{11}$. Divergent orbits have no tool at all; Tao's theorem leaves them as a set of logarithmic density zero. Conway's undecidability result for maps linear on residue classes implies that any proof must use the arithmetic of 2 and 3 specifically; the Krasikov–Lagarias system does, but only through the branching rule mod 3 and the size factors 2 and 3/2, which is why it yields exponents and not individual orbits.

Within the method, the open mathematical problem is $\lambda_k \longrightarrow 2 \iff \rho_k \longrightarrow 1$. §6 reduces it to two statements about the limit object: that the fine top-digit alignments act as independent, which the scrambling experiment supports to $2 \cdot 10^{-3}$ in λ, and that the distribution of the ×4 weight converges, which the data to k = 21 support with geometrically shrinking increments; together they would give a contraction estimate for the loss and hence $\lambda_k \longrightarrow 2$. The same reduction holds for 5x+1 with ceiling 0.6509 in place of 1, so a proof cannot rest on the contracting drift of 3x+1. We regard the data to k = 22 as consistent with that limit and not as evidence for it.

## Data and code availability

Repository: github.com/dpalmqvist/kollatz (MIT). Certificates certs/k{k}_{λ}/cert.json and certs/k{k}_{λ}_upper/ with sha256 of the vectors; independent verifier verify_cert.py; the search (kl.py, kl_mlx.py, kl_split.py); Lean modules, manifests and verifier reports in lean_formal/; measurements in rho.py, holder.py, rF2_fit.py, renorm.py, renorm2.py, spectrum.py, boundary_layer.py, defect.py, meanfield.py, weights.py, popdyn.py, kl5.py, kl5_mlx.py; the literature survey in reports/.

## References


D. Applegate, J. C. Lagarias, Density bounds for the 3x+1 problem I. Tree-search method; II. Krasikov inequalities, Math. Comp. 64 (1995) 411–426, 427–438.

D. Barina, Improved verification limit for the convergence of the Collatz conjecture, J. Supercomput. 81:810 (2025).

J. H. Conway, Unpredictable iterations, Proc. 1972 Number Theory Conf., Boulder, 49–52.

R. E. Crandall, On the "3x+1" problem, Math. Comp. 32 (1978) 1281–1292.

S. Eliahou, The 3x+1 problem: new lower bounds on nontrivial cycle lengths, Discrete Math. 118 (1993) 45–56.

C. Hercher, There are no Collatz m-cycles with m ≤ 91, J. Integer Seq. 26 (2023) 23.3.5; corrigendum 14 June 2026.

A. V. Kontorovich, J. C. Lagarias, Stochastic models for the 3x+1 and 5x+1 problems, in The Ultimate Challenge: the 3x+1 problem, AMS 2010; arXiv:0910.1944.

I. Krasikov, How many numbers satisfy the 3x+1 conjecture?, Internat. J. Math. Math. Sci. 12 (1989) 791–796.

I. Krasikov, J. C. Lagarias, Bounds for the 3x+1 problem using difference inequalities, Acta Arith. 109 (2003) 237–258; arXiv:math/0 205 002.

S. Kurtz, J. Simon, The undecidability of the generalized Collatz problem, TAMC 2007, LNCS 4484, 542–553.

J. C. Lagarias, The 3x+1 problem: an overview, in The Ultimate Challenge, AMS 2010; arXiv:2111.02635.

J. C. Lagarias, D. Weiss, The 3x+1 problem: two stochastic models, Ann. Appl. Probab. 2 (1992) 229–261.

L. Mazur, Collatz predecessor 0.90 (Lean 4 development), ProofAtlas, July 2026, commit 5f76a170.

J. W. Sander, On the (3N+1)-conjecture, Acta Arith. 55 (1990) 241–248.

J. Simons, B. de Weger, Theoretical and computational bounds for m-cycles of the 3n+1 problem, Acta Arith. 117 (2005) 51–70.

T. Tao, Almost all orbits of the Collatz map attain almost bounded values, Forum Math. Pi 10 (2022) e12; arXiv:1909.03562.

R. Terras, A stopping time problem on the positive integers, Acta Arith. 30 (1976) 241–252.

G. J. Wirsching, An improved estimate concerning 3n+1 predecessor sets, Acta Arith. 63 (1993) 205–210; The Dynamical System Generated by the 3n+1 Function, LNM 1681, Springer 1998.

E. Yolcu, S. Aaronson, M. J. H. Heule, An automated approach to the Collatz conjecture, J. Automated Reasoning 67 (2023) 15; arXiv:2105.14697.